

Joint Subcarrier Allocation and Hybrid Beamforming for THz Multiuser Communications

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Abstract—The hybrid beamforming structure is widely used in terahertz (THz) communications due to the low hardware complexity. However, its performance is severely degraded by beam squint effects in wideband THz communications resulting from the non-frequency-specific analog beamforming. To overcome this issue, in this paper, we investigate THz multiuser communications and propose a joint subcarrier allocation and hybrid beamforming (JSH) algorithm to compensate for the beam squint effects and maximize the sum-rate of users. The JSH algorithm alternates between subcarrier allocation and hybrid beamforming. Given the hybrid beamforming, the subcarrier allocation is formulated as an assignment problem and solved using the classic Hungarian algorithm. Given the subcarrier allocation, the hybrid beamforming is solved by developing a penalty-based two-loop hybrid beamforming algorithm. Simulation results show that the proposed JSH algorithm can effectively compensate for the beam squint effects and outperforms the existing ones in terms of the multiuser sum-rate.

Index Terms—Beam squint, hybrid beamforming, terahertz (THz) communications, subcarrier allocation.

I. INTRODUCTION

Terahertz (THz) communications have been prominent research topics in both academia and industry over the past few years. With abundant spectrum and high data transmission rates, they are considered key enabling technologies for the future-generation wireless communications [1].

During propagation, THz signals face severe path loss, which significantly affects the quality of service (QoS) of communications. To address this issue, massive MIMO is integrated into THz communications to mitigate the path loss by providing substantial beamforming gains. Due to the large number of antennas in massive MIMO, the efficient hybrid beamforming structure is preferred in THz communications. Although the hybrid beamforming is effective in narrow band, performance degradation arises when extending it to wideband applications due to the beam squint effects [2]. Usually, frequency-specific beamformers are required to mitigate these effects. For the hybrid beamforming structure, the phase shifter-based analog beamformer is non-frequency-specific and cannot be specifically designed for different frequencies [3]. Although the digital beamformer is frequency-specific and can be tailored for different frequencies, its design freedom is constrained by the limited number of RF chains. The disparity between the requirement for frequency-specific beamformers and the hardware limitations of hybrid beamforming structures poses significant challenges for wideband

hybrid beamforming.

Two primary approaches, namely the hardware-centric approaches [4]–[7] and the algorithm-centric approaches [8], [9], have been proposed to address the above challenges. The hardware-centric approaches focus on enhancing the hybrid beamforming structure by introducing frequency-specific analog devices, such as true-time delayers (TTDs). For instance, non-frequency-specific phase shifters are replaced by frequency-specific TTDs in [4], a delay-phase precoding (DPP) structure is proposed [5], a serial TTD-based hybrid beamforming structure is introduced [6], and dynamic subarrays with fixed TTDs are used in [7]. Although the hardware-centric approaches effectively mitigate beam squint effects, they lead to increased hardware complexity and power consumption due to the introduction of extra devices. On the other hand, the algorithm-centric approaches focus on designing analog and digital beamforming to compensate for beam squint effects without increasing the hardware complexity. For example, the analog beamforming and digital beamforming are alternately designed to approximate the fully digital (FD) beamforming in [8], and wide beams are formed to provide beamforming gains across all subcarriers in [9].

In this paper, we focus on the algorithm-centric approaches and design the hybrid beamforming for wideband THz multiuser massive MIMO. We propose a joint subcarrier allocation and hybrid beamforming (JSH) algorithm to compensate for the beam squint effects and maximize the sum-rate of users. The JSH algorithm alternates between subcarrier allocation and hybrid beamforming. Given the hybrid beamforming, the subcarrier allocation is formulated as an assignment problem and solved using the classic Hungarian algorithm. Given the subcarrier allocation, the hybrid beamforming is solved by developing a penalty-based two-loop hybrid beamforming (PT-HBF) algorithm.

Notations: Lowercase and uppercase bold symbols denote vectors and matrices, respectively. $[\mathbf{A}]_{m,:}$, $[\mathbf{A}]_{:,n}$, and $[\mathbf{A}]_{m,n}$ denote the m th row, the n th column, and the entry on the m th row and the n th column of a matrix $\mathbf{A}$. $[\mathbf{a}]_m$ denotes the m th entry of a vector $\mathbf{a}$. $|\cdot|$, $\|\cdot\|_2$, and $\|\cdot\|_F$ denote the absolute value of a scalar, the ℓ_2 -norm of a vector, and the Frobenius norm of a matrix, respectively. $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^\dagger$ denote the transpose, Hermitian transpose, and the Moore-Penrose inverse, respectively. $\mathbb{C}$ and $\mathcal{CN}(\cdot)$ denote the set of complex numbers and complex Gaussian distribution, respectively.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a wideband THz multiuser communication system, where a base station (BS) serves K users within its coverage. The BS employs an N -element uniform linear array with half-wavelength spacing to compensate for the severe path loss during propagation. For simplicity, we focus on the signal processing at the BS and assume that each user is equipped with only a single antenna.

The BS employs the orthogonal frequency division multiplexing (OFDM) with M subcarriers to support wideband communications. Let f_c and B denote the center carrier frequency and the bandwidth, respectively. Then, the carrier frequency of the m th subcarrier is given by

$$f_m = f_c + \frac{(2m-1-M)B}{2M}, \quad m = 1, 2, \dots, M. \quad (1)$$

The wide bandwidth in THz communications causes significant variations in carrier frequencies across subcarriers and results in subcarrier-dependent channel steering vectors, known as the beam squint effects [10]. Let $\alpha(N, \Psi, f_m) \in \mathbb{C}^N$ denote the channel steering vector of the m th subcarrier for the physical channel angle Ψ , where its n th entry is given by

$$[\alpha(N, \Psi, f_m)]_n = \exp\{j\pi(n-1)f_m\Psi/f_c\}. \quad (2)$$

In addition, we define $\bar{\Psi}_m \triangleq f_m\Psi/f_c$ to represent the corresponding steering channel angle for the m th subcarrier. According to the widely-adopted Saleh-Valenzuela model [11], the channel between the BS and k th user for the m th subcarrier can be modeled as

$$\mathbf{h}_{k,m} = \sum_{l=1}^{L_k} \gamma_{k,l,m} \alpha(N, \Theta_{k,l}, f_m), \quad (3)$$

where L_k denotes the number of channel paths between the BS and the k th user, $\gamma_{k,l,m}$ denotes the channel gain of the l th path for the m th subcarrier, and $\Theta_{k,l}$ denotes the physical channel angle of the l th path for the k th user.

B. Problem Formulation

In THz communications, the large number of antennas renders the classic fully digital structure impractical due to the prohibitive hardware costs. To address this issue, the efficient hybrid beamforming (HBF) structure is adopted, where the N antennas are connected to N_{RF} RF chains through an analog beamformer. Limited by the number of RF chains, the number of users simultaneously served by the hybrid beamforming cannot exceed N_{RF} . Typically, N_{RF} is small, e.g. 4 or 8, and is significantly smaller than the total number of users to be served, i.e., $N_{\text{RF}} \ll K$. This indicates that the space division multiplexing is insufficient to serve the users. In this condition, the time division multiplexing (TDM) and frequency division multiplexing (FDM) are needed. Most existing works focus on the TDM, where N_{RF} users are served in each time slot and every user can exploit the full frequency resources [5]–[7]. However, these works neglect the degrees of freedom in the frequency dimension, which may degrade the performance of hybrid beamforming. In this paper, we employ the FDM and jointly design the subcarrier

allocation and hybrid beamforming to alleviate the beam squint effects and maximize the sum-rate.

To facilitate the proposed design, the K users are assumed to be pre-divided into N_{RF} groups, where the i th group contains $\tilde{K}_i$ users and satisfies $\sum_{i=1}^{N_{\text{RF}}} \tilde{K}_i = K$. It is worth noting that the user grouping strategy is beyond the scope of this work and will be addressed in future research. Given this grouping, space division multiplexing is adopted to simultaneously serve the N_{RF} groups, while frequency division multiplexing is employed to serve users within each group. Based on (3) and the user grouping, the channel of the k th user in the i th group for the m th subcarrier can be obtained as $\bar{\mathbf{h}}_{k,i,m}$. Then, the multiuser sum-rate maximization problem can be formulated as

$$\max_{\delta_{k,i,m}, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}} \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} \sum_{k=1}^{\tilde{K}_i} \delta_{k,i,m} R(k, i, m) \quad (4a)$$

$$\text{s.t.} \quad \delta_{k,i,m} \in \{0, 1\}, \quad (4b)$$

$$\sum_{k=1}^{\tilde{K}_i} \delta_{k,i,m} = 1, \quad (4c)$$

$$\sum_{m=1}^M \delta_{k,i,m} \geq 1, \quad (4d)$$

$$\|\mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB},m}\|_F^2 \leq P, \quad (4e)$$

$$|[\mathbf{F}_{\text{RF}}]_{n,i}| = 1. \quad (4f)$$

In (4a), $R(k, i, m)$ represents the achievable rate of the k th user in the i th group using the m th subcarrier, defined as

$$R(k, i, m) \triangleq \log_2 \left(1 + \frac{G(k, i, m, i)}{\sigma^2 + \sum_{p=1, p \neq i}^{N_{\text{RF}}} G(k, i, m, p)} \right), \quad (5)$$

where $\mathbf{F}_{\text{BB},m}$ and $\mathbf{F}_{\text{RF}}$ are the digital beamformer for the m th subcarrier and the analog beamformer, respectively. In (4b), $\delta_{k,i,m}$ denotes whether the k th user in the i th group is served by the BS using the m th subcarrier. If it is served, $\delta_{k,i,m} = 1$; otherwise, $\delta_{k,i,m} = 0$. In (4c), one subcarrier is only allocated to one user in each group to support independent data transmission. In (4d), we stipulate that each user is allocated at least one subcarrier to ensure the communication QoS. In (4e), the transmit power at any subcarrier should not exceed the maximum transmit power P . In (4f), each entry in the analog beamformer should satisfy the constant modulus constraint.

III. JOINT SUBCARRIER ALLOCATION AND HYBRID BEAMFORMING ALGORITHM

In this section, we propose a joint subcarrier allocation and hybrid beamforming algorithm to solve (4).

A. Initialization

We initialize $\mathbf{F}_{\text{RF}}$ and $\mathbf{F}_{\text{BB},m}$ as

$$[\mathbf{F}_{\text{RF}}]_{:,i} = \sqrt{N} \alpha(N, \bar{\Theta}_i), \text{ and } \mathbf{F}_{\text{BB},m} = \kappa \mathbf{I}, \quad (6)$$

where $\bar{\Theta}_i$ denotes the center angle of the i th group and κ is the scaling factor to guarantee the power constraint in (4e) is satisfied.

B. Subcarrier Allocation

Given the analog and digital beamformers, the subcarrier allocation problem is formulated as

$$\begin{aligned} \max_{\delta_{k,i,m}} \quad & \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} \sum_{k=1}^{\tilde{K}_i} \delta_{k,i,m} R(k,i,m) \\ \text{s.t.} \quad & (4b), (4c), \text{ and } (4d). \end{aligned} \quad (7)$$

Note that $\delta_{k,i,m}$ for different groups are independent. Therefore (7) can be divided into N_{RF} subproblems, with the i th subproblem expressed as

$$\begin{aligned} \max_{\delta_{k,i,m}} \quad & \sum_{m=1}^M \sum_{k=1}^{\tilde{K}_i} \delta_{k,i,m} R(k,i,m) \\ \text{s.t.} \quad & (4b), (4c), \text{ and } (4d). \end{aligned} \quad (8)$$

In fact, (8) aims at allocating M subcarriers to K users, and is an assignment problem. Using the classic Hungarian algorithm [12], we can obtain the optimal solution of (8).

C. Hybrid Beamforming

Given $\delta_{k,i,m}$, each subcarrier is assigned to a corresponding user for an RF chain. We denote the assigned user channels for the m th subcarrier as $\mathbf{H}_m \in \mathbb{C}^{N \times N_{\text{RF}}}$, where

$$[\mathbf{H}_m]_{:,i} = \mathbf{h}_{k,i,m}, \text{ when } \delta_{k,i,m} = 1, \text{ for } i = 1, \dots, N_{\text{RF}}. \quad (9)$$

The optimization problem in (4) is then transformed into

$$\begin{aligned} \max_{\mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}} \quad & \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} W(m,i) \\ \text{s.t.} \quad & (4e) \text{ and } (4f), \end{aligned} \quad (10)$$

where

$$W(m,i) \triangleq \log_2 \left(1 + \frac{|[\mathbf{H}_m]_{:,i}^H \mathbf{F}_{\text{RF}} [\mathbf{F}_{\text{BB},m}]_{:,i}|^2}{\sigma^2 + \sum_{p=1, p \neq i}^{N_{\text{RF}}} |[\mathbf{H}_m]_{:,i}^H \mathbf{F}_{\text{RF}} [\mathbf{F}_{\text{BB},m}]_{:,p}|^2} \right). \quad (11)$$

To solve (10), we first introduce auxiliary variables $\mathbf{F}_m$ defined as

$$\mathbf{F}_m \triangleq \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB},m}. \quad (12)$$

Accordingly, the sum-rate in (11) is rewritten as

$$\bar{W}(m,i) \triangleq \log_2 \left(1 + \frac{|[\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,i}|^2}{\sigma^2 + \sum_{p=1, p \neq i}^{N_{\text{RF}}} |[\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,p}|^2} \right). \quad (13)$$

The power constraint in (4e) can be rewritten as

$$\|\mathbf{F}_m\|_{\text{F}}^2 \leq P. \quad (14)$$

Thus, (10) is reformulated as

$$\max_{\mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}} \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} \bar{W}(m,i) \quad (15a)$$

$$\text{s.t.} \quad (4f), (12), \text{ and } (14). \quad (15b)$$

According to [13], any nontrivial stationary point of $\mathbf{F}_m$ maximizing $\bar{W}(m,i)$ satisfies the power constraint in equality. Based on this property, we can drop the power constraint in

(14) and convert (15) into an equivalent form expressed as

$$\max_{\mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}} \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} \bar{W}(m,i) \quad (16a)$$

$$\text{s.t.} \quad (4f), \text{ and } (12), \quad (16b)$$

where $\bar{W}(m,i)$ is defined as

$$\bar{W}(m,i) \triangleq \log_2 \left(1 + \frac{|[\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,i}|^2}{\frac{\sigma^2 \|\mathbf{F}_m\|_{\text{F}}^2}{P} + \sum_{p=1, p \neq i}^{N_{\text{RF}}} |[\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,p}|^2} \right). \quad (17)$$

Note that the solution of (15) can be obtained by multiplying the corresponding solution of (16) with a scaling factor to guarantee the power constraint is satisfied with equality. The optimization in (16) is challenging because of the nonconvex log-sum objective in (16a), nonconvex constant modulus constraint in (4f), and nonconvex equality constraint in (12). In the following, we address the challenging objective and constraints to solve (16).

We first address the nonconvex log-sum objective. According to [14], (16a) can be converted to

$$\min_{v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}} \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} w_{m,i} e_{m,i} - \log w_{m,i}, \quad (18)$$

where

$$\begin{aligned} e_{m,i} \triangleq \quad & \sum_{p=1, p \neq i}^{N_{\text{RF}}} |v_{m,i} [\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,p}|^2 \\ & + |v_{m,i} [\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,i} - 1|^2 + \frac{|v_{m,i}|^2 \sigma^2 \|\mathbf{F}_m\|_{\text{F}}^2}{P}, \end{aligned} \quad (19)$$

and $v_{m,i}$ is the receiving factor of the i th data stream for the m th subcarrier.

The nonconvex equality constraint in (12) presents significant challenges to the optimization of (15). To simplify the problem, we employ the penalty method to remove the constraint and integrate it into the objective as

$$\begin{aligned} \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}) \\ \triangleq \sum_{m=1}^M \sum_{i=1}^{N_{\text{RF}}} w_{m,i} e_{m,i} - \log w_{m,i} + \frac{1}{\rho} \|\mathbf{F}_m - \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB},m}\|_{\text{F}}^2, \end{aligned} \quad (20)$$

where ρ is the penalty factor. Thus, (15) can be reformulated as

$$\begin{aligned} \min_{v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}} \quad & \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}) \\ \text{s.t.} \quad & (4f). \end{aligned} \quad (21)$$

Note that (21) is a multivariable programming problem, and is challenging to solve. We then develop a PT-HBF algorithm to decompose it into several tractable subproblems.

In the outer loop of the PT-HBF algorithm, we initialize ρ as a large value to relax the hardware constraint of the hybrid beamforming and ensure a good value of sum-rate. Then, we gradually decrease the value of ρ by multiplying it with a factor β less than one to ensure the hardware constraint of hybrid beamforming is ultimately satisfied.

In the inner loop of the PT-HBF algorithm, we fix ρ and use

Algorithm 1 Joint Subcarrier Allocation and Hybrid Beamforming (JSH) Algorithm

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1: Input:  $N, M, N_{\text{RF}}, \bar{h}_{k,i,m}, \beta, \rho$ , and  $P$ .
2: Initialization: Initialize  $\mathbf{F}_{\text{RF}}$  and  $\mathbf{F}_{\text{BB},m}$  via (6).
3: while stop conditions in Sec. III-D are not satisfied do
4:   /*Perform Subcarrier Allocation*/
5:   Obtain  $\delta_{k,i,m}$  by solving (23).
6:   /*Perform Hybrid Beamforming*/
7:   while outer-loop stop conditions are not satisfied do
8:      $\rho \leftarrow \rho \cdot \beta$ .
9:     while inner-loop stop conditions are not satisfied do
10:      Obtain  $v_{m,i}$  via (23).
11:      Obtain  $w_{m,i}$  via (25).
12:      Obtain  $\mathbf{F}_m$  via (29).
13:      Obtain  $\mathbf{F}_{\text{RF}}$  via (34).
14:      Obtain  $\mathbf{F}_{\text{BB},m}$  via (36).
15:     end while
16:   end while
17: end while
18: Output:  $\delta_{k,i,m}, \mathbf{F}_{\text{RF}}$  and  $\mathbf{F}_{\text{BB},m}$ .

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the alternating minimization method to optimize the variables involved in (21). Then, we introduce how these variables are optimized separately.

1) *Optimization of $v_{m,i}$:* When fixing the other variables, the optimization of (21) can be converted to

$$\min_{v_{m,i}} \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}). \quad (22)$$

Note that (22) is an unconstrained optimization problem. Letting $\partial \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}) / \partial v_{m,i} = 0$, we can express the solution of (22) as

$$v_{m,i} = \frac{[\mathbf{F}_m]_{:,i}^H [\mathbf{H}_m]_{:,i}}{\sum_{p=1}^{N_{\text{RF}}} |[\mathbf{H}_m]_{:,i}^H [\mathbf{F}_m]_{:,p}|^2 + \frac{\sigma^2 \|\mathbf{F}_m\|_F^2}{P}}. \quad (23)$$

2) *Optimization of $w_{m,i}$:* When fixing the other variables, the optimization of $w_{m,i}$ can be converted to

$$\min_{w_{m,i}} \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}). \quad (24)$$

Note that (24) is also an unconstrained optimization problem. Let $\partial \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}) / \partial w_{m,i} = 0$, then the solution of (24) can be expressed as

$$w_{m,i} = 1/e_{m,i}. \quad (25)$$

3) *Optimization of $\mathbf{F}_m$:* When fixing the other variables, the optimization of $\mathbf{F}_m$ can be converted to

$$\min_{\mathbf{F}_m} \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}). \quad (26)$$

Note that (26) is an unconstrained problem and can be solved via gradient-based methods. Define $\mathbf{f}_{m,i} \triangleq [\mathbf{F}_m]_{:,i}$. Then, we have

$$\mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}) \\ \propto \sum_{p=1}^{N_{\text{RF}}} w_{m,p} |v_{m,p}|^2 \left(|[\mathbf{H}_m]_{:,p}^H \mathbf{f}_{m,i}|^2 + \frac{\sigma^2 \|\mathbf{f}_{m,i}\|_2^2}{P} \right) \\ - 2w_{m,i} \mathcal{R}\{v_{m,i}^* [\mathbf{H}_m]_{:,i}^H \mathbf{f}_{m,i}\} + \frac{1}{\rho} \|\mathbf{f}_{m,i} - \mathbf{F}_{\text{RF}}[\mathbf{F}_{\text{BB},m}]_{:,i}\|_2^2, \quad (27)$$

where $\propto$ denotes the operation of proportionality by omitting the irrelevant items. Then, we have

$$\frac{\partial \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m})}{\partial \mathbf{f}_{m,i}} \\ = \sum_{p=1}^{N_{\text{RF}}} w_{m,p} |u_{m,p}|^2 \left([\mathbf{H}_m]_{:,p} [\mathbf{H}_m]_{:,p}^H \mathbf{f}_{m,i} + \frac{\sigma^2 \mathbf{f}_{m,i}}{P} \right) \\ - v_{m,i} w_{m,i} [\mathbf{H}_m]_{:,i} + \frac{1}{\rho} (\mathbf{f}_{m,i} - \mathbf{F}_{\text{RF}}[\mathbf{F}_{\text{BB},m}]_{:,i}). \quad (28)$$

Letting $\partial \mathcal{L}(v_{m,i}, w_{m,i}, \mathbf{F}_m, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB},m}) / \partial \mathbf{f}_{m,i} = 0$, we can obtain the optimal solution of $\mathbf{f}_{m,i}$ as

$$\mathbf{f}_{m,i} = (\Psi^H \Psi)^{-1} \Psi^H \boldsymbol{\eta}, \quad (29)$$

where

$$\Psi \triangleq \sum_{p=1}^{N_{\text{RF}}} w_{m,p} |u_{m,p}|^2 \left([\mathbf{H}_m]_{:,p} [\mathbf{H}_m]_{:,p}^H + \frac{\sigma^2}{P} \mathbf{I} \right) + \frac{\mathbf{I}}{\rho}, \\ \boldsymbol{\eta} \triangleq v_{m,i} w_{m,i} [\mathbf{H}_m]_{:,i} + \frac{1}{\rho} \mathbf{F}_{\text{RF}}[\mathbf{F}_{\text{BB},m}]_{:,i}. \quad (30)$$

Stacking $\mathbf{f}_{m,i}$ in (29), we can obtain the optimal solution of $\mathbf{F}_m$ in (26).

4) *Optimization of $\mathbf{F}_{\text{RF}}$:* When fixing the other variables, the optimization of $\mathbf{F}_{\text{RF}}$ can be converted to

$$\min_{\mathbf{F}_{\text{RF}}} \sum_{m=1}^M \|\mathbf{F}_m - \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB},m}\|_F^2 \\ \text{s.t. (4f)}. \quad (31)$$

Note that (31) is a least-squares problem with constant modulus constraints. Following [6], we convert (31) to a tractable form expressed as

$$\min_{\mathbf{F}_{\text{RF}}} \sum_{m=1}^M \left\| \mathbf{F}_m (\mathbf{F}_{\text{BB},m})^\dagger - \mathbf{F}_{\text{RF}} \right\|_F^2 \\ \text{s.t. (4f)}. \quad (32)$$

Note that

$$\sum_{m=1}^M \left\| \mathbf{F}_m (\mathbf{F}_{\text{BB},m})^\dagger - \mathbf{F}_{\text{RF}} \right\|_F^2 \\ \propto -2\mathcal{R} \left\{ \text{Tr} \left\{ \sum_{m=1}^M \left(\mathbf{F}_m (\mathbf{F}_{\text{BB},m})^\dagger \right) \mathbf{F}_{\text{RF}}^H \right\} \right\}. \quad (33)$$

Therefore, the optimal solution of (32) can be expressed as

$$\mathbf{F}_{\text{RF}} = \exp \left\{ j \mathcal{A} \left\{ \sum_{m=1}^M \left(\mathbf{F}_m (\mathbf{F}_{\text{BB},m})^\dagger \right) \right\} \right\}. \quad (34)$$

where $\mathcal{A}\{\cdot\}$ denotes the angles of complex numbers.

5) *Optimization of $\mathbf{F}_{\text{BB},m}$:* When fixing the other variables, the optimization of $\mathbf{F}_{\text{BB},m}$ can be converted to

$$\min_{\mathbf{F}_{\text{BB},m}} \sum_{m=1}^M \|\mathbf{F}_m - \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB},m}\|_F^2. \quad (35)$$

Note that (35) is an unconstrained least-squares problem, and its optimal solution can be expressed as

$$\mathbf{F}_{\text{BB},m} = (\mathbf{F}_{\text{RF}}^H \mathbf{F}_{\text{RF}})^{-1} \mathbf{F}_{\text{RF}}^H \mathbf{F}_m. \quad (36)$$

Now, we discuss the stop conditions of the PT-HBF algo-

rithm. The outer loop aims at guaranteeing the solution of the hybrid beamformers approaches that of the FD beamformers. Therefore, it stops when the discrepancy of the two beamformers is less than a predefined threshold ϵ , i.e., $\|\mathbf{F}_m - \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB},m}\| \leq \epsilon$. The inner loop aims at maximizing the multiuser sum-rate. Therefore, it stops when the discrepancy of the objectives between two adjacent iterations is less than a predefined threshold ϵ .

D. Stop Conditions of the AM-JSH Algorithm

We iteratively perform the subcarrier allocation in Section III-B and the hybrid beamforming in Section III-C until one of the following two stop conditions is satisfied. i) The predefined maximum number of iterations is reached. ii) The disparity of the objective between adjacent two iterations is less than a predefined threshold ϵ .

Finally, we summarize the developed JSH algorithm in **Algorithm 1**.

IV. SIMULATION RESULTS

Now, we evaluate the performance of the proposed JSH algorithm for wideband multiuser THz massive MIMO with the HBF structure. The number of antennas is set to $N = 64$ and the center carrier frequency is set to $f_c = 120$ GHz. The channel between the BS and the user for each subcarrier consists of one dominant path and two non-dominant paths, where their channel gains follow $\mathcal{CN}(0, 1)$ and $\mathcal{CN}(0, 0.01)$, respectively. The noise power is normalized to $\sigma^2 = 1$. The number of RF chains in the HBF structure is set to $N_{\text{RF}} = 4$.

To simplify the expression, we refer to the proposed algorithm as the “*HBF (JSH)*” in simulation results. In addition, we include the following five algorithms to provide a comparative analysis.

- **FD (JSD)**: This scheme performs the joint subcarrier allocation and digital beamforming (JSD) algorithm based on the FD structure, which can be obtained by removing the hardware constraints of the HBF structure in (4f) and (12), and performing the same procedures as the JSH algorithm.
- **DPP (JSH)**: This scheme performs the JSH algorithm based on the DPP structure, where the number of TDDs for each RF chain is four and the maximum time delay of the TTD is 67 ps.
- **FD (TDB)**: This scheme performs the TDM-based digital beamforming (TDB) based on the FD structure.
- **DPP (THB)**: This scheme performs the TDM-based hybrid beamforming (THB) based on the DPP structure via the method in [15].
- **HBF (THB)**: This scheme performs the TDM-based hybrid beamforming (THB) based on the HBF structure via the method in [15].

Since the *FD (JSD)*, *DPP (JSH)*, and *HBF (JSH)* are based on the FDM, we refer to them as FDM-based algorithms. On the other hand, the *FD (TDB)*, *DPP (THB)*, and *HBF (THB)* are based on the TDM, we refer to them as TDM-based algorithms.

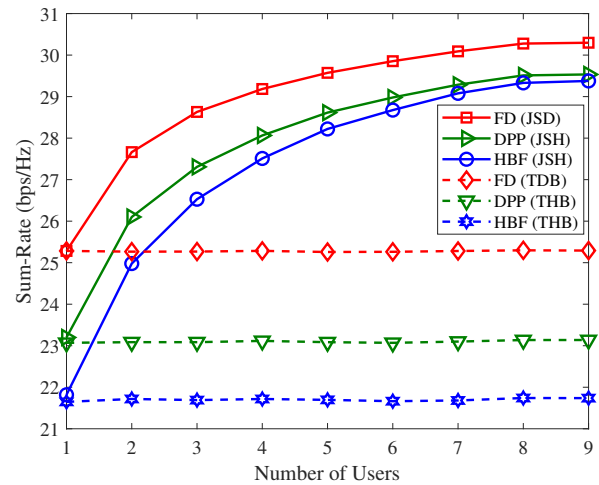


Fig. 1. Comparisons of different schemes for varying numbers of users in the clustered user scenarios.

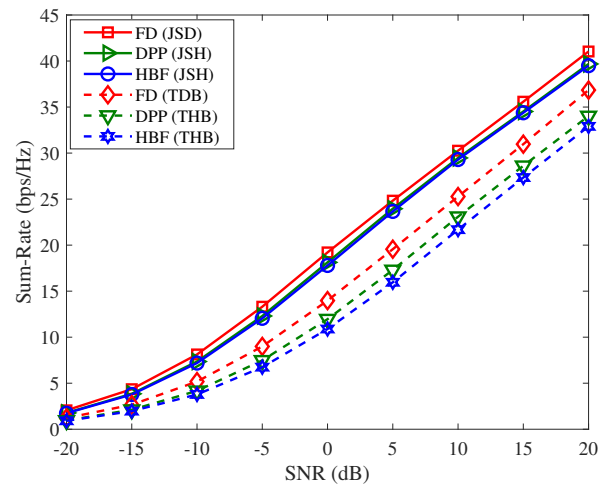


Fig. 2. Comparisons of different schemes for varying SNRs in the clustered user scenarios.

In Fig. 1, we compare different algorithms in terms of the sum-rate for varying numbers of users in each group, where the x-axis denotes the number of users in each group. The signal-to-noise ratio (SNR), the number of subcarriers, and the bandwidth are set to be 10 dB, 11, and 12 GHz, respectively. From the figure, the FDM-based algorithms have similar performance as the TDM-based algorithms when the number of users is one. As the number of users increases, the performance of the FDM-based algorithms increases while the performance of the TDM-based algorithms remains unchanged, leading to the increasing advantages of the FDM-based algorithms. This is because the FDM-based algorithms can select the subcarriers with better channel quality for communications while the TDM-based algorithms have to use all the subcarriers regardless of their channel quality. In addition, the performance gap between the HBF structure

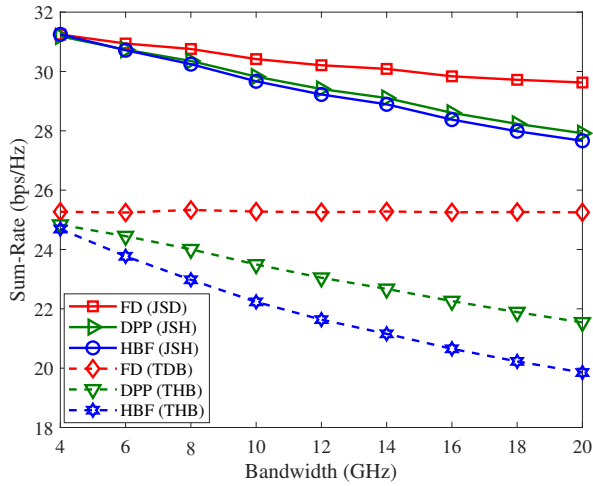


Fig. 3. Comparisons of different schemes for varying bandwidths in the clustered user scenarios.

and the FD structure for the FDM-based algorithms is much smaller than that of the TDM-based algorithms. This is because the FDM-based algorithms can dynamically allocate subcarriers to users according to the matches between the analog beamforming and channels of users across different subcarriers, which cannot be achieved with the TDM-based algorithms. Moreover, the performance gap between the HBF structure and the FD structure for the FDM-based algorithms decreases with the increase in the number of users due to the increased freedom in subcarrier allocation. Specifically, the performance of the *HBF (JSH)* can closely approach that of the *DPP (JSH)* when the number of users is larger than six, which indicates that the HBF structure can achieve good performance without the need to introduce TDDs in the considered scenarios.

In Fig. 2, we compare different algorithms in terms of the sum-rate for varying SNRs, where we set the number of users, the number of subcarriers, and the bandwidth as 8, 11, and 12 GHz, respectively. From the figure, the FDM-based algorithms outperform the TDM-based ones for all SNRs. In addition, with the JSH algorithm, the performance gap between the HBF structure and the FD structure as well as the DPP structure can be effectively reduced for all SNRs.

In Fig. 3, we compare different algorithms in terms of the sum-rate for varying bandwidths, where we set the number of users, the SNR, and the number of subcarriers as 8, 10 dB, and 11, respectively. From the figure, as the bandwidth increases, the performance of the HBF structure and the DPP structure decreases due to the beam squint effects. However, the FDM-based algorithms decrease more slowly than the TDM-based algorithms, verifying the effectiveness of the proposed scheme in alleviating the beam squint effects.

V. CONCLUSION

In this paper, we have investigated multiuser THz communications and propose a JSH algorithm to compensate for

the beam squint effects and maximize the sum-rate of users. The JSH algorithm alternates between subcarrier allocation and hybrid beamforming. Given the hybrid beamforming, the subcarrier allocation has been formulated as an assignment problem and solved using the classic Hungarian algorithm. Given the subcarrier allocation, the hybrid beamforming has been solved by developing a penalty-based two-loop hybrid beamforming algorithm. Simulation results have shown that the proposed JSH algorithm can effectively compensate for the beam squint effects and outperforms the existing works in terms of the multiuser sum-rate.

VI. ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China under Grants 624B2035 and U22B2007.

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